

Abhinav Chinnusamy

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EDUCATION

University of Wisconsin-Madison

Master of Science in Electrical and Computer Engineering

Aug 2024 – May 2026

Coursework: Power Electronic Circuits, Power Electronics Laboratory, Solid State Power Conversion, Introduction to Optimization, Electric Machines, Electric Machines Laboratory, Advanced Computer Control of Machines and Processes

Indian Institute of Technology

Bachelor of Technology in Electrical Engineering

Dharwad, India

Aug 2020 – Apr 2024

RESEARCH EXPERIENCE

Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC)

Madison, WI

Graduate Researcher

Aug 2024 – Present

- Impedance matching for pulse generators.
- Validating microstrip designs.

Power and Energy Group, IIT Dharwad

Dharwad, India

Undergraduate Researcher

Dec 2022 – June 2024

- Developed an 800V, 70A SiC solid-state circuit breaker for EVs and DC homes under Dr. Satish Naik.
- Designed PCBs and performed DPTs for GaN half-bridge inverters in MMC applications under Dr. Abhijit Kshirsagar.

ENGINEERING EXPERIENCE

Annapurna Labs (Amazon Web Services)

Austin, TX

Hardware Development Engineer Intern

June 2025 – August 2025

- Developed tools to enhance debugging processes for accelerator card diagnostics.
- Analyzed and repaired RMA cards, identifying and resolving hardware bugs.
- Validated high-density capacitors and supported next-generation GPU card bring-up.

PUBLICATIONS

1. **A. Chinnusamy**, H. Baik, and J. Roy, “High Power Density Solid State Marx Generator Based Pulsed Power Driver for Nonlinear Transmission Line,” in *2025 IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia)*, Bengaluru, India, 2025, pp. 1–6, doi: 10.1109/ECCE-Asia63110.2025.11112355.
2. D. Dsa, **A. Chinnusamy**, S. N. Banavath, and E. L. Carvalho, “Implementation of Protection Features for a Modular Bidirectional Solid-State Battery Disconnecter,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 6, pp. 6001–6010, Nov. 2024.
3. **A. Chinnusamy**, D. Dsa, and S. N. Banavath, “Intelligent Battery Protection System for Electric Vehicle Applications,” in *2024 IEEE 18th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG)*, Poland, June 2024, pp. 1–6.